

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools. (BL4, BL5)

Assessment Tool Weightage: 20%

Questions: 40 Total Marks: 40

Annexure A **Mock Interview**

Code of Conduct:

I S. Karthikeyan / 192419048 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Section A — Analytical & Evaluative Questions

Q1. Analyze how network congestion affects packet delay and packet loss. (BL4)

When traffic entering a link exceeds its capacity, packets queue up in router buffers, so each packet waits longer before transmission — this is queuing delay, the main contributor to increased end-to-end delay under congestion. Once buffers fill completely, incoming packets have nowhere to go and are dropped, causing packet loss. So congestion first shows up as rising delay, and if it isn't relieved, escalates into loss, which then forces retransmissions and worsens the congestion further — a feedback loop.

Q2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)

A star topology depends entirely on the central switch/hub: if it fails, the whole network goes down, even though a single cable fault only isolates one node. A mesh topology has multiple redundant paths between nodes, so if one link or even one node fails, traffic can be rerouted through an alternate path — giving it much higher fault tolerance and reliability, at the cost of far more cabling and configuration complexity than a star.

Q3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)

Every router a packet passes through adds its own processing delay (reading the header and deciding where to forward it), queuing delay (waiting behind other packets), and transmission delay (pushing the packet onto the next link). End-to-end delay is the sum of these delays across every hop, so more routers simply mean more hops, and each additional hop adds its own delay components on top of the ones before it.

Q4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)

Voice and video are delay- and jitter-sensitive but can tolerate very little loss before quality visibly degrades, unlike bulk file transfers which tolerate delay but need reliability. QoS mechanisms (traffic classification, prioritized queuing, bandwidth reservation) give voice/video packets priority over less time-sensitive traffic, reducing their queuing delay and jitter so calls and streams stay smooth even when the network is under load from other applications.

Q5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)

Fiber carries signals as light rather than electrical pulses, giving it far higher bandwidth capacity, much lower attenuation over distance, and complete immunity to electromagnetic interference compared to copper — all of which support significantly higher achievable throughput and longer cable runs without repeaters. The trade-off is higher material and installation cost and the need for specialized termination skills, so fiber is generally justified for backbone/high-speed links rather than every last short connection.

Q6. Analyze how network segmentation using VLANs improves performance and security. (BL4)

A VLAN splits one physical network into multiple logical broadcast domains, so broadcast traffic from one department's devices no longer floods every port on the switch — this directly reduces unnecessary traffic and improves performance. Security improves because devices in different VLANs cannot communicate directly without going through a router/firewall enforcing access policy, containing the impact of a compromised device to its own segment.

Q7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)

TCP adds reliability through handshaking, acknowledgements, and retransmission of lost segments, which guarantees delivery but adds overhead and can introduce delay when retransmissions occur — well suited to file transfer, email, and web traffic where correctness matters more than instant delivery. UDP has no such overhead and simply sends packets, accepting some loss in exchange for lower, more predictable latency, which is why it's preferred for VoIP and live video where a late retransmitted packet is less useful than a dropped one.

Q8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)

Without load balancing, traffic can concentrate on one server or link while others sit underused, creating a bottleneck that increases delay and risks overload even though total network capacity is sufficient. Distributing requests across multiple servers/links keeps utilization more even, improving throughput and response time, and if one server/link fails, traffic can continue flowing through the remaining ones — improving both performance and availability simultaneously.

Q9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)

Frequent routing updates let a network adapt quickly to topology or link changes, keeping paths accurate and avoiding stale routes that could cause loops or blackholes. However, each update itself consumes bandwidth and router processing time, so if updates are too frequent relative to how often the topology actually changes, the overhead can itself reduce the bandwidth available for real traffic — so update frequency is a genuine efficiency trade-off, not a purely ‘more is better’ setting.

Q10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)

Firewalls inspect every packet against rule sets and encryption requires computational work to encrypt/decrypt each packet, both of which add processing delay and can reduce effective throughput, especially under deep packet inspection or high encryption strength. The trade-off is generally justified because the cost of a security breach (data loss, downtime, reputational damage) usually far outweighs the performance overhead, so the practical answer is to right-size security to the sensitivity of the traffic rather than maximizing either performance or security in isolation.

Section B — Problem-Solving & Scenario-Based Questions

Q11. A network has high delay — how will you identify and fix the issue? (BL4)

I'd first isolate which delay component is dominant by running ping/traceroute to see where round-trip time jumps between hops, then check link utilization and router CPU/queue statistics on the suspect hop. If it's queuing delay from congestion, I'd apply QoS prioritization or add capacity/load-balance traffic; if it's a specific overloaded or misconfigured router, I'd address that device directly rather than treating it as a whole-network problem.

Q12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)

Available bandwidth is a ceiling, not a guarantee — actual throughput can be held down by high packet loss forcing constant TCP retransmission and window shrinking, high latency limiting how much data a TCP window can have in-flight, congestion at a downstream bottleneck link, protocol/hardware duplex mismatches, or simply too many devices sharing that bandwidth simultaneously.

Q13. How would you reduce packet loss in a congested network? (BL4)

I'd start by increasing effective capacity where feasible (upgrading the bottleneck link or load-balancing across multiple paths), then apply QoS so critical traffic is prioritized before buffers overflow, and consider traffic shaping/rate-limiting non-critical bulk traffic at the source so it doesn't overwhelm the link in the first place rather than only reacting after loss occurs.

Q14. A VoIP call is breaking frequently — how would you troubleshoot? (BL4)

I'd check jitter and packet loss statistics on the path first, since VoIP is especially sensitive to both; if jitter is high I'd verify QoS prioritization is actually applied to the voice traffic's DSCP/port range, and if packet loss is high I'd look for a congested or failing link along the path. I'd also confirm the jitter buffer settings on the VoIP endpoint itself, since an undersized buffer will drop late-arriving packets even on an otherwise healthy link.

Q15. Given a network with high jitter, suggest improvements. (BL4)

I'd apply QoS to give delay-sensitive traffic a dedicated, prioritized queue so its delay stays consistent rather than varying with bulk-traffic bursts, reduce the number of hops/queuing points on that path where possible, and increase jitter-buffer size on receiving endpoints to smooth out remaining variation — while also checking for a specific congested link that could be the root cause rather than a network-wide issue.

Q16. How will you simulate a congested network in Packet Tracer? (BL5)

I'd build a topology where multiple end devices send continuous or high-volume traffic (e.g. via generated pings or traffic generators) through a single shared, lower-bandwidth link, then use Packet Tracer's simulation mode to watch queuing build up on that link's interface and observe rising delay/dropped packets in the event list, which reproduces congestion behaviour in a controlled, observable way.

Q17. If packets are delayed in a router, what steps will you take? (BL4)

I'd check the router's CPU and interface queue utilization first to see if it's a processing bottleneck or a link-congestion bottleneck, review the routing table for inefficient or flapping routes causing reprocessing, and check for misconfigured QoS/ACLs adding unnecessary per-packet inspection overhead. Depending on the cause, the fix is either offloading load (better routing/QoS), or hardware/link capacity upgrade.

Q18. How would you optimize a network for online gaming? (BL4)

Online gaming needs low and, more importantly, consistent latency rather than raw bandwidth, so I'd prioritize game traffic with QoS, minimize the number of hops/routing complexity on the path, use a wired connection instead of Wi-Fi where possible to avoid wireless jitter, and ensure no other device on the network is saturating the shared link with bulk downloads during play.

Q19. Design a small network and explain how you will measure its performance. (BL5)

For a small office, I'd design a star topology with a core switch connecting department switches and a router/firewall to the internet, with VLANs separating departments. To measure performance I'd track throughput (bits transferred per second on key links), latency and jitter (via ping/traceroute to internal and external targets), and packet loss (via extended ping tests or SNMP interface error counters), comparing these against baseline figures taken when the network was first deployed.

Q20. If RTT is very high, how will you diagnose the issue? (BL4)

I'd run traceroute to see exactly which hop introduces the largest jump in round-trip time, since that isolates the problem to a specific link or device rather than the whole path. If the jump appears at the last hop before the destination, the issue is likely local to that server/link; if it's consistent across many destinations from the same starting point, the issue is more likely with my own outbound link or gateway.

Section C — Communication & Presentation Questions

Q21. Explain network performance to a non-technical person. (BL5)

Think of a network like a highway system: throughput is how many cars can get through per minute, delay is how long each car's journey takes, jitter is how unpredictable that journey time is from one car to the next, and packet loss is like cars that never arrive at all. 'Good network performance' just means cars (your data) move quickly, predictably, and reliably reach their destination.

Q22. Describe throughput using a real-life example. (BL5)

Throughput is like the actual number of customers a coffee shop serves per hour, versus its theoretical maximum capacity (bandwidth) if every seat and barista worked perfectly. Even if the shop *could* serve 100 customers an hour, a slow register or crowded entrance might mean it only actually serves 60 — that real, achieved number is throughput.

Q23. Present how packet loss affects video streaming. (BL5)

When packets carrying video data are lost, the player either has to wait for a retransmission (causing buffering/freezing) or simply skip the missing data (causing visible glitches, blurring, or frozen frames), depending on the protocol used. Even a small percentage of sustained packet loss is noticeable to viewers, which is why streaming services invest heavily in adaptive bitrate and redundant delivery paths to keep loss as close to zero as possible.

Q24. Explain delay types clearly in under 1 minute. (BL4)

There are four main delay types on any hop: propagation delay, the time for a signal to physically travel the link; transmission delay, the time to push all the packet's bits onto the link; queuing delay, the time a packet waits in a buffer before its turn; and processing delay, the time a router takes to examine the packet and decide where to send it. End-to-end delay is simply the sum of all four, across every hop on the path.

Q25. Describe Cisco Packet Tracer in a professional way. (BL5)

Cisco Packet Tracer is a network simulation and visualization tool that lets students and professionals design, configure, and test network topologies — routers, switches, end devices, and cabling — in a risk-free virtual environment. It supports both real-time and simulation modes, the latter letting you step through individual packets hop-by-hop, which makes it particularly useful for learning and demonstrating protocol behaviour without needing physical hardware.

Q26. Explain QoS in simple and technical terms. (BL5)

Simple version: QoS is like a priority lane that lets time-sensitive traffic (a video call) go ahead of traffic that can wait (a file download), so the call doesn't stutter just because someone else is downloading a large file. Technical version: QoS classifies and marks traffic (e.g. via DSCP values), then applies queuing and scheduling policies at each network device so higher-priority classes receive preferential bandwidth, lower delay, and lower drop probability than best-effort traffic.

Q27. How would you present your simulation results to a client? (BL4)

I'd lead with the headline finding in plain language (e.g. 'the network can reliably support 50 concurrent video calls before quality degrades'), then back it up with a clear chart of the key metric trends (throughput, delay, loss) rather than raw logs, and close with a concrete recommendation tied to their business need — avoiding jargon unless the client's technical team specifically asks for the underlying detail.

Q28. Explain jitter using a real-world analogy. (BL5)

Jitter is like a bus that's scheduled to arrive every 5 minutes, but sometimes shows up in 2 minutes and sometimes in 9 — the average interval might still be 5 minutes, but the inconsistency makes it hard to rely on.

In a network, that inconsistency in packet arrival timing is exactly what causes choppy audio or video even when average delay looks fine.

Q29. Describe how you tested network performance in a lab. (BL4)

In our Packet Tracer/lab exercises, I built a small topology, then used extended ping and traceroute to record baseline latency and packet loss between endpoints, generated sustained traffic to observe throughput under load, and compared these readings before and after introducing a deliberate bottleneck (e.g. a lower-bandwidth link) to see how the metrics changed — which helped confirm the cause-and-effect relationships between topology choices and measured performance.

Q30. Explain the importance of performance metrics in one structured answer. (BL5)

Performance metrics matter because they turn ‘the network feels slow’ into something measurable and fixable: throughput tells you actual achieved capacity, delay and jitter tell you how usable the network is for real-time applications, and packet loss tells you about reliability. Together, they let an administrator diagnose the specific cause of a problem, verify whether a fix actually worked, and plan capacity before users notice degradation — rather than reacting only after complaints arrive.

Section D — Personal & Behavioural Questions

The following are first-person template answers. Personalize the specifics (projects, experiences) to your own background before the actual interview.

Q31. Introduce yourself and explain your knowledge of network performance. (BL5)

I'm S. Karthikeyan, currently studying Computer Networks as part of my engineering coursework, with hands-on practice in Cisco Packet Tracer building and testing topologies. I've worked with core performance metrics — throughput, delay, jitter, and packet loss — both theoretically and by simulating congestion and measuring how these metrics respond, which has given me a practical sense of how design choices translate into real performance outcomes.

Q32. Why are you interested in networking and simulation tools? (BL5)

I find networking interesting because it sits at the intersection of theory and very tangible, observable behaviour — you can design a topology, simulate real traffic through it, and directly see the effect of your design choices on delay and throughput. Simulation tools like Packet Tracer make that feedback loop fast and safe, which is what first drew me into wanting to understand networks more deeply.

Q33. What makes you confident in analyzing network performance? (BL5)

I'm comfortable connecting a metric to its underlying cause rather than just naming the metric — for example, explaining why high jitter often traces back to inconsistent queuing rather than treating it as an unexplained symptom. That comes from repeatedly working through scenario-based problems and simulations rather than just memorizing definitions.

Q34. Explain a concept you are very strong in (related to networking). (BL5)

I'm particularly comfortable with the relationship between congestion, delay, and packet loss — how queuing builds up as a link approaches capacity, how that shows up first as rising delay, and how sustained congestion eventually forces packet drops once buffers saturate. I can walk through that chain of cause and effect clearly, and connect it to practical fixes like QoS or load balancing.

Q35. How would you handle a question you don't know? (BL5)

I'd say honestly that I'm not certain, then reason through it out loud from what I do know — for example, relating an unfamiliar term back to a related concept I understand well — rather than guessing confidently or staying silent. I think showing a clear thought process matters more than pretending to know something I don't.

Q36. Describe a situation where you solved a technical problem. (BL4)

During a lab exercise, a simulated network was showing much lower throughput than expected despite adequate bandwidth. I traced it back to a duplex mismatch between two connected interfaces by checking the interface statistics, corrected the configuration, and confirmed throughput returned to expected levels — which taught me to check configuration details rather than assume a capacity problem first.

Q37. Why should we select you for a networking role? (BL5)

I bring a solid grounding in performance metrics and how they connect to real design decisions, hands-on simulation experience, and a habit of diagnosing problems methodically rather than guessing — I focus on understanding the 'why' behind a metric, which I think translates directly into being able to troubleshoot real network issues effectively.

Q38. What challenges did you face while learning Packet Tracer? (BL5)

Initially, connecting simulation-mode packet traces back to the underlying theory (why a packet took a particular path, or why delay showed up where it did) took practice — the tool shows you what happened, but

interpreting why required working through several examples deliberately. Repeating exercises with intentional faults (like a deliberately misconfigured link) helped build that intuition.

Q39. How do you improve your technical skills regularly? (BL5)

I work through hands-on exercises rather than relying only on reading theory, deliberately try to break and then fix simulated setups to build diagnostic intuition, and revisit concepts I found difficult until I can explain them simply — which is usually a good test of whether I actually understand something.

Q40. Demonstrate confidence while explaining a complex concept simply. (BL5)

Take TCP congestion control: put simply, when a connection first starts, it sends a little data, and if that arrives fine, it sends a bit more next time — gradually ramping up until it either finds the network's true capacity or starts seeing loss, at which point it backs off and starts ramping up again. It's a constant, self-adjusting feeling-out process, which is exactly why TCP performance can vary even on the same network from one moment to the next.